TheoryBERT: Comprehensive Technical Project Report

BERT language analysis, formal correctness and input-specific visualization for theory of computation and compiler education

Report date: 29 October 2026
Scope: local implementation, dataset provenance, model experiments, web application, formal algorithms and reproducibility.
Companion: an eight-page IEEE-style research manuscript with a component architecture diagram.

Executive Overview

TheoryBERT interprets English formal-language requirements with BERT and presents executable mathematical objects. The DFA workspace supports machine construction, state/transition inspection, test-string simulation and exact comparison when an independent reference exists. The CFG workspace predicts grammar productions, parses inputs and checks derivation trees. The compiler workspace connects a predicted arithmetic grammar to lexical tokens, a parse tree, an AST and symbolic intermediate code.

The central engineering principle is to keep language interpretation, formal execution and visual representation consistent. A neural prediction is recorded as a concrete object. Mathematical checks operate on that object. The browser and optional video renderer visualize its actual transitions or productions. The system does not use a fixed illustrative graph as a substitute for the prediction, and it does not treat a successful animation as proof of correct interpretation.

The newest experiment trains a 537,385-parameter semantic-plan head over frozen evaluated BERT features. It predicts typed operations and learns to copy literal bit patterns. Exact constructors compile the plan into a DFA. It passed every promotion gate and became the local default. The final active local DFA approach is the BERT semantic-plan decoder. Presentation components remain unchanged.

The report includes negative and limited results. Generated tests are easier than some fresh wording. The robust direct fine-tune was rejected. The grammar model performs strongly on its generated canonical-production test but remains imperfect on a small fresh challenge. These facts define the evidence, rather than being hidden behind aggregate scores.

Prepared from the local source tree and recorded experiment artifacts. This document is not an IEEE publication. Public source links identify original research or external artifacts; local generated datasets are not represented as publicly hosted releases.

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Measured: a result recorded by a completed run. Implemented: an executable code path exists. Regression: a previously inspected case or test. Proposed: future work without a completed experiment. Verified: a specific formal claim supported by the appropriate checker. These terms are deliberately not interchangeable.

Dataset versions overlap. SHA-256 identifies exact bytes, not data quality.

Problem, Objectives and Requirements

The Problem Being Solved

An English request describes a set of strings, while a DFA describes that set through finite memory and transitions. Students must infer which distinctions between prefixes matter, preserve enough context for later input and handle boundary cases such as the empty word. For grammars, a learner must connect recursive productions with actual derivations. Compiler exercises add lexical categories, precedence, associativity and intermediate representations. These representations are related, but none is interchangeable with a generic diagram.

The original practical concern was that the same visualization appeared for unrelated requests. A correct solution requires the mathematical content to change with the language and the execution content to change with the test input. The graph should not gain arbitrary states merely because a longer test string was entered. It should gain states when the represented language requires more distinguishable histories. This distinction guides both the architecture and the test criteria.

Functional Requirements

Requirement	Concrete acceptance condition
BERT language analysis	Learned English requests invoke a trained BERT encoder and task head.
Relevant DFA	Displayed states, accepting roles and transitions equal the current candidate object.
String-specific execution	Input edits produce a new trace and acceptance result for that machine.
Grammar and compiler	Predicted productions support checked parses; AST and IR match the accepted expression.
Honest verification	Reference-free predictions remain previews; exact equivalence requires an independent DFA.
Preserve presentation	The new experiment changes model logic without changing final diagram styles or playback.

Nonfunctional Requirements

The available computer has 8 GB of memory and constrained disk space. The experiment must avoid parallel heavy training, unnecessary backbone duplication and indefinitely retained feature caches. Resource snapshots and stopping thresholds must accompany training. Loading a different task head should reuse the same immutable encoder when its fingerprint matches. The implementation should retain previous evaluated checkpoints for rollback without copying the base model into every new head directory.

The scientific objective is improved semantic correctness under a controlled comparison. Parameter count, training time and response latency are measured separately. The reporting objective is a complete, reproducible account of sources, data filtering, architecture, tests and limitations. The reports must not imply universal language understanding, general CFG equivalence or a complete industrial compiler.

Literature and Technical Foundations

BERT provides contextual language representations that can be adapted to supervised tasks [1]. TheoryBERT uses the pretrained base uncased model as an English encoder, then predicts formal outputs with explicit task heads. The pretrained model card is the source of its artifact lineage and inherited pretraining information [10]. Downstream automata performance is measured in this project rather than inferred from BERT's general NLP results.

Kushman and Barzilay study natural-language regular-expression generation using semantic unification [2]. Their work highlights the value of semantic equivalence when distinct formal expressions mean the same thing. Locascio et al. study learned translation from paired descriptions and regular expressions [3]. Their repository supplies the KB13 mirror used by this project [9]. The local filtered binary task differs from the original source task, so published scores are not used as direct baselines.

The exBERT work studies domain-vocabulary adaptation under limited compute [4]. The repository contains an inspired extension with added embedding rows and a residual contextual layer. Its location and training condition differ from the original architecture, and the V3 experiment also fine-tunes two BERT blocks. The new plan head freezes that V3 encoder. The report keeps these conditions distinct.

Herzig et al. examine intermediate representations for compositional semantic parsing [5]. The present plan experiment applies the general design motivation to automata. Oren et al. analyze contextual representations, alignment and program-template balance [8]. These findings motivate literal-copy supervision and operator-specific analysis here. The implementation does not reproduce either paper's full model, data or experimental controls.

PICARD demonstrates formal constraints during autoregressive output generation [6]. TheoryBERT instead predicts plan positions in parallel and uses a small typed prefix beam afterward. The shared idea is formal admissibility during decoding; no PICARD code or SQL parser is used. A typed output remains capable of expressing the wrong meaning, so structural validity is evaluated separately.

CheckList motivates targeted behavioral tests beyond aggregate held-out accuracy [7]. The local challenge suites probe wording, boundaries, negation and composition. Cornell's compiler notes support the classical connection from regular specifications to total automata and lexical processing [12]. NLTK supplies the Earley parser used for context-free derivations [11]. These are established foundations, not new formal algorithms introduced by this project.

The project-specific research question is whether a compact, typed, copy-capable semantic target improves a BERT-based educational application under local compute limits. A stronger causal claim would require matched training data, matched capacities, several seeds and independent human-authored evaluation. None of the cited papers makes those missing controls unnecessary.

Research Positioning and Design Decisions

Why BERT Remains Central

The user requirement is that English interpretation happen through BERT analysis. In learned mode, both the direct and plan predictors tokenize the complete requirement and run the trained encoder. The plan model learns operator selection, composition and numeric arguments from contextual features. It learns where to copy a literal. No inference-time rule maps a phrase such as ending with directly to a suffix constructor. A separately labelled rule baseline exists for comparison, not as a hidden production fallback.

Why Formal Algorithms Are Still Necessary

The neural system supplies an interpretation. Formal tools supply exact semantics for the predicted representation. Complementing a complete DFA, constructing an intersection or checking a derivation is mathematical execution. Asking the neural model to guess every transition is one possible representation, but it is not required to keep BERT responsible for English understanding. The plan path retains the neural interpretation boundary and makes later construction deterministic.

A useful example is the request to accept words containing 00101 and having an even number of ones. The predicted plan should retain a literal 00101 and a count-modulo condition, joined by conjunction. Its compiler builds the two component languages, explores their reachable product and minimizes it. If BERT predicts an odd count instead, the compiler faithfully constructs the wrong language. An independent reference can expose that error; successful construction alone cannot.

Why the Existing Visualization Is Preserved

The final visual representation is a public contract already capable of displaying arbitrary supported DFA JSON. The new work changes how that object is produced. It keeps graph layout, SVG styling, trace animation and optional Manim scenes stable. This makes the comparison easier to review and avoids confusing model improvements with presentation changes. Correctness is checked by comparing the visible graph and trace with the generated object.

What Is Unique at the Project Level

The distinctive combination is a BERT-centered interpretation path, explicit verification provenance, state-dependent and input-dependent educational views, grammar and compiler integration, versioned local datasets and resource-bounded experiments. This is an integrated engineering contribution. The report makes no claim that typed semantic parsing, neural copying, automata products or constrained decoding were invented here.

The experiment also treats deployment selection as part of correctness. A candidate can improve a fresh subset while harming known product behavior. Promotion is therefore gated on fresh validation, established validation and product fixtures. A negative promotion result is preserved as a useful experiment, and the candidate remains available by explicit checkpoint selection.

System Component Architecture

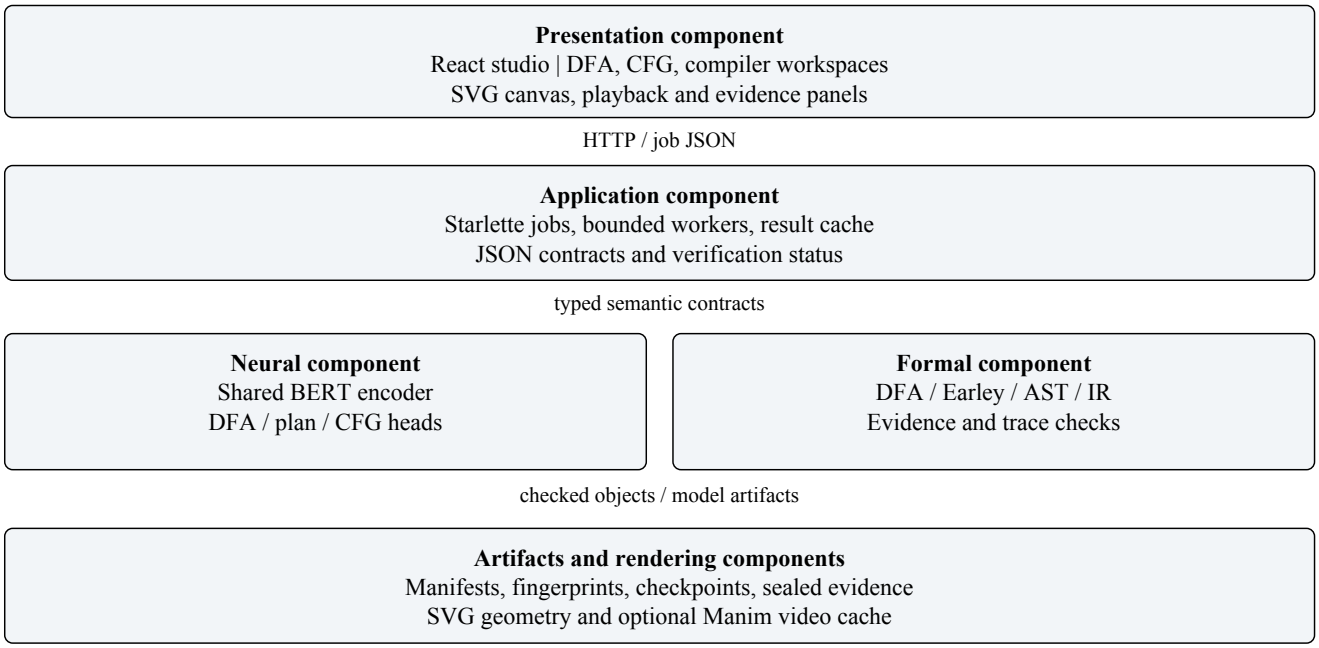


Figure 1. Component/deployment architecture and named formal interfaces.

Component	Responsibilities and contracts
React presentation	Description and input editors; settings; graph/tree/AST playback; evidence and research panels.
Starlette application	Validates request shape, resolves active model, manages bounded jobs, retains result/evidence records.
Neural subsystem	Shared immutable encoder; direct DFA head; grammar token decoder; typed plan and literal-pointer head.
Formal kernel	DFA JSON checks, construction, simulation, minimization, equivalence; CFG parsing and tree checks; AST and IR consistency.
Artifact store	Pinned source metadata, dataset manifests, model configs, hashes, training logs and evaluation outputs.
Rendering subsystem	Graph/tree geometry, browser SVG and optional sealed-evidence Manim exports.

The main interfaces are HTTP job JSON between browser and application, typed DFA/grammar/plan objects between model and formal kernel, and checked objects plus traces between formal kernel and rendering. The component view intentionally avoids procedural decision diamonds or an implied single execution sequence. Optional preview, manual-entry and reference-verified paths use the same mathematical contracts with different provenance.

A head checkpoint names the encoder path and SHA-256 fingerprint. Loading rejects a mismatch. Model caching shares the compatible encoder between plan and grammar tasks, while each task retains its own small head. This avoids a second full BERT copy for an ordinary task switch. The application remains the authority for verification status and renderable evidence.

Direct BERT DFA Baseline

Representation and Training Targets

The direct model predicts a complete binary DFA with at most eight states and canonical start state q0. Its pooled contextual representation drives three heads: state-count logits, accepting-state logits and transition-destination logits for every state and binary symbol. Targets are reachable, minimal automata renamed in deterministic breadth-first order. This reduces arbitrary state-name variation in supervised learning.

State count uses cross-entropy. Accepting roles use binary cross-entropy on active states. Transition destinations use cross-entropy on active source rows, with destinations outside the gold active range masked during training. Padded rows do not contribute. At inference, each destination is restricted to the predicted active-state range. Structural completeness therefore follows from the output contract, while correct language semantics remain learned.

Model Lineage

The local direct experiments begin from google-bert/bert-base-uncased. The current V3 training condition fine-tunes the upper two pretrained encoder blocks and trains the contextual extension and structured heads. Lower blocks and protected embeddings remain frozen. Training logs compare fingerprints before and after fine-tuning and assert that frozen weights are unchanged. The condition has 19,015,056 trainable parameters.

V3 uses 6,275 supervised descriptions, maximum sequence length 64, batch size two, gradient accumulation four, seed 42, learning rate 8e-6 for the updated BERT blocks and 8e-5 for the other trainable parameters. Its recorded MPS training time is 1,553.0 seconds. Its best original validation equivalence is 90.34%. The model directory records the exact dataset fingerprint and pretrained revision.

Candidate Search and Verification

The first candidate uses argmax state count and local decisions. Bounded alternatives explore uncertain count, accepting-bit and transition choices through score-gap search. Candidates are deduplicated; a reference is not consulted to create them. When a reference exists, the formal verifier can stop at the first equivalent candidate. Without a reference, the product returns the first prediction as an explicitly unverified preview.

The direct representation is flexible for small arbitrary transition systems but has many dependent graph decisions. Adding a state changes the set of valid destinations and the interpretation of several rows. Literal identity, pattern overlap and Boolean composition must all be encoded implicitly in predicted tables. This is the motivation for testing a semantic target, rather than assuming that another similar direct fine-tune will solve the structural difficulty.

The baseline is retained for matched comparison and rollback. Its eight-state bound is part of its task definition. Targets requiring more states must not be counted as ordinary supported failures without separately identifying the capacity limitation.

New BERT Semantic-Plan Model

Architecture

The new head has 537,385 trainable parameters. It uses immutable token features from the evaluated V3 encoder, projects width 768 to 128 and applies two Transformer decoder layers. Each layer has four attention heads, feed-forward width 256 and dropout 0.1. Twenty-four learned queries represent maximum output positions. A linear head predicts typed prefix tokens, and two projections supply start/end pointer scores over input tokens.

The model is parallel rather than autoregressive: every output query attends to contextual input memory, and query self-attention lets output positions interact. The head does not consume a handcrafted English parse. Its loss supplies supervision for operator identity, symbolic arguments, output termination and copied literal boundaries. Encoder computation uses inference mode and does not update any BERT parameter.

Plan Vocabulary and Scope

Operators	Meaning
ALL, NONE	Accept every binary word or no binary word.
STARTS, ENDS, CONTAINS, EXACT	Prefix, suffix, contiguous substring or one complete literal word.
NOT, AND, OR	Complement, intersection and union of regular languages.
COUNT_EQ / LE / GE	Equality or inclusive bound on occurrences of one symbol.
COUNT_MOD	A symbol count leaves a given remainder modulo a positive modulus.
LENGTH_EQ / LE / GE / MOD	The analogous total-length requirement.

The typed schema distinguishes subexpressions, patterns, symbols and integer arguments. Implementation validation permits literals of length 1 through 32, integers 0 through 16, positive moduli with smaller nonnegative remainders, bounded nesting and at most 64 final DFA states. The training catalogue uses patterns through length eight and moduli 2 through 8. Extrapolation to the full implementation bounds has not been established.

Supervision and Objective

Targets are prefix programs with an EOS marker and padding. Single-bit patterns use learned fixed classes. Longer literals use COPY plus start/end token labels. The loss is token cross-entropy plus 0.6 times start-pointer cross-entropy plus 0.6 times end-pointer cross-entropy at copy positions. Padded token positions and non-copy pointer positions are excluded. Invalid or unaligned supervision raises an error rather than being silently repaired.

The encoder fingerprint binds the head to the exact V3 weights used for feature extraction. A different encoder requires another compatible training/evaluation condition. The small checkpoint stores head weights and configuration without copying the full backbone.

Literal Copying and Typed Decoding

Why Literal Copying Matters

Binary patterns can have leading zeros, repeated fragments and lengths unseen in a particular training subset. A model that chooses from a fixed catalogue of literal strings cannot faithfully support new literals. The pointer head instead scores a start token and an end token, allowing the predicted COPY action to reproduce the exact original substring. The interpretation of what the pattern means still comes from learned operator tokens.

Candidate span extraction checks token offsets and raw characters. A valid copied pattern is a complete contiguous binary span, not a subword cut from the middle of a larger bit literal. It preserves the original text exactly, including leading zeros. This is a type and boundary constraint, not an English keyword rule. The neural pointers choose among available literals when a request contains several of them.

Bounded Prefix Search

A beam item records accumulated score cost, predicted prefix values and a stack of argument types still required. At an expression position it permits the supported operators; at a pattern position it permits single-bit classes and COPY when a valid span exists; symbol and number positions have their respective finite vocabularies. Composite operators push their argument types. A completed stack requires an EOS prediction before the tree is accepted.

Each state expands at most three typed token alternatives. COPY expands at most two highest-scoring spans. Beam width is eight and maximum plan length is 24. The search limits compositional nesting, and the compiler independently validates the completed tree. Scores use gaps relative to the strongest token at each position and relative span scores, avoiding an unexamined preference for short one-state plans. Equivalent completed DFAs are deduplicated.

Failure and Abstention Behavior

The compiler rejects an invalid modulus, a remainder outside its range, unsupported arguments, excessive construction size or excessive final state count. If no valid plan remains within the bounded search, the predictor reports an error. It does not fall back to phrase matching or retrieve a benchmark answer. A syntactically valid but semantically wrong plan remains a wrong prediction and is scored accordingly.

The decoder receives no gold DFA during inference. The independent reference is only available to evaluation or the application verifier after candidates are produced. This boundary prevents reference-assisted construction from inflating first-candidate accuracy. Within-five success is intentionally a separate, reference-assisted metric.

Example Contract

For a predicted AND(CONTAINS("00101"), COUNT_MOD("1", 2, 0)), the formal compiler must preserve the literal and enforce an even count of ones. Unit tests inject ideal neural logits to verify exact copying and tree reconstruction independently of training. Separate predicate tests enumerate short words to check that compilation matches conjunction, negation, counts and length conditions. These tests validate implementation mechanics, not the trained model's understanding of arbitrary English.

Formal DFA Algorithms and Guarantees

Structural Validation and Simulation

A DFA requires a nonempty unique state set, explicit alphabet, declared start and accepting states and exactly one declared destination for every state/symbol pair. Strict JSON loading rejects duplicate keys before they could conceal contradictory rows. Simulation follows transitions from the start state and records each consumed symbol, source and target. The final accepting status is computed from the machine, not inferred from the animation.

Construction from Plans

Pattern constructors encode relevant suffix/prefix memory, including overlap. Exact words and count/length bounds use the supported regular-expression compiler. Modular requirements use residue states. Complement changes accepting roles on a complete DFA. Intersection and union explore reachable pairs of component states and apply the corresponding Boolean accepting condition. An intermediate 256-pair construction budget and a 64-state final budget constrain the new plan path.

The regex compiler has a deliberately supported dialect and explicit repetition/state limits. It is not a drop-in implementation of every PCRE feature. The public dataset import therefore filters unsupported constructs and checks accepted binary restrictions with bounded word comparisons. Formal source labels and filtering decisions are retained for audit.

Reachability, Minimization and Canonicalization

Reachability removes states inaccessible from the start. Partition refinement begins with accepting/nonaccepting groups and repeatedly separates states by destination-group signatures. Canonicalization traverses the minimal reachable DFA in fixed alphabet order and assigns q0, q1 and subsequent names. This yields a stable language identity for equivalent complete binary DFAs. The implementation is a straightforward refinement procedure, not a claimed optimal Hopcroft implementation.

Exact Equivalence and Counterexamples

Product BFS starts from the candidate/reference start pair. If their accepting memberships differ, the predecessor map reconstructs a shortest distinguishing input. Otherwise it explores successor pairs until exhausted or a product budget is reached. Exhaustion without disagreement establishes exact equivalence over the matching alphabet. Empty input is represented by an empty symbol list; it must not be confused with absence of a counterexample.

For complete machines with matching alphabets, exploration is bounded by the product of state counts times alphabet size. Minimization and a readable graph are useful analysis operations but do not establish equivalence to English. A formal reference can be supplied manually or by an independent benchmark fixture. Without it, the system can validate and simulate the candidate while withholding an equivalence claim.

The same minimized object feeds graph geometry, transition tables, access strings, completion strings and trace inspection. Keeping these views derived from one object prevents inconsistencies between a state counter and the actual canvas.

External Data and Pretrained Sources

Pretrained Encoder Artifact

Source: google-bert/bert-base-uncased on Hugging Face

The model revision is 86b5e0934494bd15c9632b12f734a8a67f723594. The local source manifest records download metadata and SHA-256 values for artifact files. The pretrained weights file fingerprint is 68d45e234eb4a928074dfd868cead0219ab85354cc53d20e772753c6bb9169d3. The model artifact is Apache-2.0 licensed. The project reuses its contextual English knowledge and adapts it through downstream supervised tasks.

The model card identifies BookCorpus and English Wikipedia as original pretraining sources. They are inherited lineage, not locally downloaded training datasets in this project. This report therefore does not add billions of pretraining words to the supervised project-data count or imply that the team performed large-scale BERT pretraining. Source-card limitations and biases also remain relevant to downstream interpretation.

KB13 Description/Regex Pairs

Source: pinned DeepRegex KB13 mirror

The downloaded source and target files contain 824 aligned rows. The local import accepts 55 after binary-alphabet restriction and supported-dialect checking. It rejects 497 unsupported-dialect rows, 271 constant languages after binary restriction and one excessive repeat bound. Subsequent corpus deduplication leaves 47 public-regex-labelled descriptions in the V3 version. Fifty-five imported rows and 47 retained rows describe different stages and are not contradictory.

The supplied datasets/license.md is an MIT license with 2016 Nick Locascio copyright. Its notice must accompany redistributed source material as required by that license. The report links the actual pinned files in the source appendix. The project does not claim to train on the full KB13 benchmark or the larger DeepRegex synthetic corpus. Other popular regex/semantic-parsing datasets are literature context only unless explicitly listed as a local input.

Filtering and Audit Limits

Restricting a regex to binary strings can substantially change its language. A source expression about letters may become empty, universal or otherwise irrelevant after restriction. Constant-language removal limits this artifact. Supported conversions are checked against Python regex behavior on short binary words, but bounded import checks are not presented as a proof of equivalence for all regex dialect features. Unsupported features are rejected instead of guessed.

The local manifest records exact URLs, byte lengths and hashes for descriptions, regex targets and license. Generated corpus manifests then record the resulting rows and filtering counts. This two-level provenance distinguishes the external bytes from the project's derived supervision.

DFA Corpus Lineage and Use

Local dataset	Rows / languages	Train / validation / test
data/examples/sample.jsonl	261 / 96	181 / 39 / 41
english-dfa.jsonl	1,785 / 208	1,363 / 213 / 209
english-dfa-expanded.jsonl	3,207 / 208	2,437 / 391 / 379
english-dfa-v3.jsonl	6,275 / 696	4,889 / 745 / 641
english-dfa-robust.jsonl	6,738 / 1,468	4,924 / 1,173 / 641
english-plan-v1.jsonl	11,415 / 2,914	9,187 / 1,171 / 1,057

Table 1. Generated DFA versions live under data/generated/ unless a path is shown. They overlap.

The shipped sample supports baseline mechanics, structural examples and canonical language splitting. English DFA v1 adds practical semantic templates and the supported KB13 subset. The expanded version increases English supervision without increasing the 208-language identity set. V3 adds 3,068 examples and broadens the target set to 696 languages. Its 6,275 rows are the supervised basis for the evaluated direct V3 model.

V3 family counts are 1,351 suffix, 1,338 prefix, 1,325 substring, 864 count-modulo, 504 exact-literal, 424 length-modulo, 236 bounded-count, 180 avoids, 47 public-regex and six edge-case descriptions. These counts describe rows rather than balanced language families. Several descriptions can express the same canonical language. A high row-weighted result can therefore differ from a language-balanced result.

The robust direct candidate uses a merged 6,738-row version with additional wording and protected challenge-related language groups. Its evaluation failed promotion gates; that dataset is still part of the project experiment history. Its test partition retains the 641-case V3 regression set. The new training file is not evidence that every extra row improved the active system.

The plan version annotates 6,274 of the 6,275 V3 descriptions with an equivalent copyable typed plan. One description has no supported copyable annotation. Additional plan-generated descriptions and eligible retired-development rows yield 11,415 total examples and 2,914 canonical languages. Only the 9,187 training rows contribute gradients; 1,171 validation rows select the head. The 1,057 plan-file test rows are retained for a separate audit and are not the frozen v2 comparison reported here.

A plan label is accepted because its compiled DFA has the same canonical identity as the gold DFA and its required multi-bit literal appears in the description. This is offline supervision generation. At inference, the model does not search that catalogue for an English match. Many formal plans can denote one language; label selection favors a shorter stable equivalent representation, while evaluation remains language based.

Local generation code supplies the construction recipe. Generated JSONL files are ignored by Git. The companion dataset archive supplies the actual local corpus versions, grammar training files, pinned KB13 input/license and a complete file/hash register. Public repository links do not claim that these local artifacts are hosted remotely.

Grammar and Challenge Datasets

Dataset / role	Examples	Partition counts
Original grammar training corpus	1,299	999 train, 170 val, 130 test
Robust grammar augmentation	456	371 train, 85 validation
Robust merged grammar corpus	1,652	1,267 train, 255 val, 130 test
Mixed challenge v1	165	71 dev, 46 val, 48 test
Plan comparison v2	254	134 validation, 120 test
Shipped manual challenge fixtures	3	challenge only

The original grammar generator covers matching counts, balanced delimiters, ordered blocks, palindromes and arbitrary words over selected symbols from a, b, x, y, 0 and 1. It also includes canonical arithmetic grammars as known fixtures. Its 1,299 examples contain 123 unique production identities. The model is a learned production-token decoder, not a retrieval table of English patterns.

The robust augmentation contributes 456 descriptions. Merging includes deduplication and removal of protected training identities, so 1,299 plus 456 is not the final row count. The robust dataset has 1,652 rows and 119 grammar identities. Arithmetic, ambiguous addition and compiler grammars are intentionally known training fixtures. Their product behavior is tested, but it is not held-out grammar discovery.

The original grammar dataset is stored at artifacts/bert-grammar/dataset.jsonl. A pooled-feature pilot dataset also exists at artifacts/bert-grammar-pilot/dataset.jsonl. The robust merged file is artifacts/grammar-robust/dataset.jsonl. The augmentation is data/generated/grammar-robust.jsonl. These are project-generated corpora without a separate external dataset license or public download URL.

Mixed challenge v1 contains 100 DFA, 38 CFG and 27 compiler requests. Its reserved test contains 23 DFA, 14 CFG and nine compiler cases. It distinguishes wording interpretation, structure generation, unsupported state bounds and downstream visualization. It has now been inspected and partly used through eligible development cases, so future work should treat it as retired regression evidence rather than an untouched test.

Plan comparison v2 contains 254 unique formal language identities outside the new plan training split. Its descriptions use a distinct wrapper and selected held-out wording styles. Validation has 134 cases; test has 120. Targets include cases above the direct head's capacity, so the analysis also reports a matched subset with at most eight states and no more than 64 baseline tokens. The file is frozen by SHA-256 before either model predicts it.

The small shipped challenge file has three manually selected demonstrations, excluded from the original sample splits. A separate suffix-01 JSON object is an independent reference fixture, not another training corpus. Public links for generator code and the exact local data register appear later in this report.

Splits, Leakage and Provenance

Canonical Language Grouping

DFA examples are grouped by the canonical minimized reachable machine before splitting. All paraphrases of an equivalent language stay together. This is stronger than random description splitting, which can place the same mathematical answer in both training and test. The plan dataset inherits all known prior language roles and assigns genuinely new identities by a deterministic hash rule. Assertions reject a language identity that appears in multiple partitions.

An early plan-data build failed because a retired development language with a newly assigned held-out role could be added to training. The build stopped before writing the corpus. A shared role-resolution function now governs generated rows and development additions. The rebuilt dataset passes a per-language partition assertion, and unit tests verify that no comparison language is in plan training. This is a corrected construction issue, not a completed invalid training condition.

What Is Still Shared

Language-group protection does not imply unseen linguistic constructions. Generated families share operator concepts, vocabulary and some template patterns. The v2 wrapper differs from training descriptions, while its underlying clause families remain generated. The evaluated V3 encoder has its own supervised history, but old held-out formal language groups remain excluded from its gradients. The new comparison is designed as a local controlled test, not a claim of independently sampled real-world requests.

Exact duplicate descriptions are removed when building the plan corpus. Multi-bit targets must be copyable from the description. New comparison descriptions are checked against training descriptions, and their canonical language identities are checked against the training set. The comparison manifest records exact bytes. A second experiment should create another comparison version rather than overwrite v2 after observing errors.

Selection and Test Boundaries

Internal validation chooses the best head checkpoint by plan-and-copy exactness, with loss as a tie-break. Decoded original and fresh validation then apply the frozen deployment gates. Only after the selection record is written does the comparison script score the reserved test partition. The test is not used for retraining or changing the promotion criteria in this experiment.

The V3 test has already been evaluated across project iterations and is now a regression benchmark. The mixed challenge test is also already inspected. Results from those sets remain useful, but repeated reuse reduces their independence. A report must identify that status rather than repeatedly call the same known set unseen.

Artifact Identity

Each important corpus has a SHA-256 fingerprint. A head config records its dataset fingerprint and exact encoder-weight fingerprint. Active-model pointers name selected artifacts. Evaluation outputs include per-case results and protocol metadata. Hashes prevent silent substitution of bytes; they do not establish absence of every conceptual leakage source. Reproducibility additionally requires source version, dependency versions, random seeds and access to the local generated artifacts.

Training and System Resource Management

New Experiment Configuration

The completed run uses seed 64, AdamW with learning rate 0.001 and weight decay 0.001, batches of 32 for the head, two CPU threads and gradient clipping at norm one. Feature extraction runs in batches of eight. The maximum configured input is 96 tokens, and training explicitly rejects overlength descriptions. The head has 537,385 parameters. The frozen encoder is reused from artifacts/bert-v3/best; no new backbone-weight copy is written.

Training allows up to 60 epochs and stops after ten stale validation epochs. Internal validation selection maximizes exact program tokens together with copied-literal correctness, then minimizes validation loss. The best head is saved independently of optimizer state. Test examples are excluded from feature extraction and checkpoint selection. The training summary reports 10,358 processed train/validation records, while the complete corpus has 11,415 rows including its untouched audit-test split.

The completed feature-extraction and head-training time is 230.8 seconds. Best internal validation plan-and-copy exactness is 94.19%. This is not decoded language equivalence. Relative to 19,015,056 trainable parameters in direct V3 training, the new head reduces trainable capacity by 97.17%. The full encoder still participates in inference, so total deployed model size does not shrink by that percentage.

Packed Feature Cache

The first extraction attempt allocated a padded dense feature tensor. It was stopped after memory pressure increased, before head training. The revised implementation writes only real contextual token rows to a packed float16 NumPy memory map. Prefix-sum boundaries locate each example. Head batches load slices and pad only to their local maximum length. The temporary cache is deleted after successful completion; an interrupted cache can be identified by its run directory rather than by deleting unrelated files.

Resource Guard and Operational Limits

The new run logs disk availability, system memory availability and swap usage. It reserves the expected cache size before writing and enforces at least 3 GiB free disk, at least 8% available system memory and no more than 0.75 GiB swap growth relative to run start. Checks occur during extraction and periodically during head training. The run executes at reduced scheduling priority. Only one heavy training or evaluation job is run at a time.

At the recorded training completion snapshot, free disk was 3.69 GiB and memory availability was 34%. Resource snapshots concern that moment, not a permanent guarantee. Background applications and the operating system can change free space independently. No user files, prior datasets or unrelated processes are deleted or terminated to manufacture a successful run.

The new head occupies roughly 2.1 MB in its local checkpoint directory. Broader project storage is dominated by historical encoder checkpoints and the Python environment. Training-time memory and CPU peak utilization are not presented as professionally instrumented energy or power measurements.

Evaluation and Promotion Protocol

Primary and Secondary Metrics

Metric	Meaning
First-candidate equivalence	Exact DFA language equality of the first prediction; failures count as incorrect.
Within-five success	At least one of five neural candidates matches a supplied independent reference.
Structural validity	A complete and well-formed predicted automaton, regardless of intended meaning.
Plan-and-copy exactness	Teacher-target program-token and literal exactness during head selection.
Grammar production exactness	Equality of the predicted canonical production token output, not general CFG equivalence.
Pipeline success	Task-specific expected outcome, including correct syntax rejection for negative compiler cases.
Prediction latency	Tokenization, encoder, head, candidate decoding and formal construction time.

First-candidate equivalence is the main semantic measure. A valid machine can still be wrong, and a correct candidate found only with a gold reference is not an unaided first answer. Released correctness is a property of the verification gate because it uses the same independent reference as evaluation. It must not be advertised as 100% model accuracy.

Frozen Gates

The protocol is written to artifacts/plan-promotion-protocol.json before model predictions on the comparison. A candidate must improve matched fresh validation first-candidate equivalence by at least ten percentage points, retain at least 88.34% on the 745-case original validation set and pass every known DFA product fixture. Matched cases require at most eight target states and no more than 64 baseline tokenizer tokens.

The selection script evaluates baseline and candidate validation, writes a selection record, then evaluates the reserved fresh test and the old regression test. The active pointer changes only if every validation/product gate passes. The candidate is not retrained after test exposure. The script refuses to overwrite an existing selection file, making accidental repeated selection against the same frozen test harder.

Interpretation Limits

The new and direct systems differ in architecture, output representation and training supervision. The comparison evaluates the full new local approach. It cannot isolate representation as the only cause of improvement. Timing is sequential on the same machine with two CPU threads, but background load, cache warmth and Python overhead remain. There is one seed, no confidence calibration and no independent human evaluation. These limitations are part of the reported result.

Established Experiments and Their Limits

Recorded experiment	Outcome	Interpretation
Direct V3, 641-row test	562 first correct; 599 within five	87.68% / 93.45% on generated regression corpus.
Earlier matched 379-row comparison	62.53% to 90.24% first correct	An expansion comparison on another set, not the 641-row result.
Robust direct candidate	Not promoted	Original validation regression and product gate failures.
Generated robust grammar test	130/130 exact productions	Templated grammar corpus; arithmetic fixtures are in training.
Mixed test CFG	3/14 to 10/14	Small fresh wording set, now retired.
Mixed test compiler	0/9 to 9/9 pipeline success	Includes expected syntax rejection as well as valid expressions.

The active direct baseline before this experiment is V3, not the rejected robust checkpoint. V3 training uses 6,275 descriptions and 696 language identities. Its original validation first-equivalence is 90.34%. The rejected robust fine-tune had improved some fresh validation wording but scored only 87.38% on the original validation set in the recorded gate comparison and failed known fixtures. It therefore did not become the product default.

The mixed challenge reserved test has 23 DFA requests, of which seven are correct under the direct model. Two targets require nine states and exceed the original eight-state output space. The small subset score illustrates a broad wording gap; it should not be conflated with V3's generated-test score. Subsequent plan training uses eligible retired-development data only, so v1 is no longer described as an untouched evaluation source.

The grammar decoder was improved from an initial feature-pooling pilot to a cross-attention production decoder and then a robust wording condition. The pilot and final results are separate model conditions. The robust model's generated-test exactness is strong, but CFG challenge accuracy of 10/14 is a more cautious indicator of wording sensitivity. Exact canonical productions also do not measure all semantically equivalent alternative grammars.

The old rule baseline is intentionally limited to a few phrase patterns. Its semantic coverage on the shipped test is small. It serves as an explicit experimental comparator, not the mechanism behind learned product outputs. The tiny random-BERT smoke experiment checks training and checkpoint mechanics only. It is never represented as a meaningful pretrained-language model.

The vanilla versus adapted frozen-backbone study, three initialization strategies across three seeds, a matched-capacity adapter control, DistilBERT and a paid external LLM baseline are not completed accuracy experiments. Configuration files and optional adapters do not constitute results. The project reports what was run and leaves these conditions as future work.

New Plan Comparison and Final Selection

Test metric	Direct V3	Semantic plan
Fresh cases	120	120
Fresh first correct	4/120 (3.33%)	86/120 (71.67%)
Fresh within five	5.83%	99.17%
Matched cases	57	57
Matched first	7.02%	66.67%
Matched within five	12.28%	100.00%
Original 641 first	87.68%	95.32%
Original within five	93.45%	97.66%
Extra-capacity first	0.00%	76.19%
Fresh median seconds	0.03400	0.03966
Original median seconds	0.03774	0.04881

The full fresh set includes 63 cases outside the matched direct-model scope. Improvements on those cases partly reflect the new formal output capacity. The 57-case matched subset provides the fairer supported-scope comparison. The large difference on that subset is promising, but its small generated sample and shared operator catalogue limit generalization claims.

Matched fresh validation improves by 51.35 percentage points. Candidate original validation is 98.39%. Product fixtures pass: true. The candidate passed every promotion gate and became the local default. The selection is recorded before fresh test evaluation, and the active pointer preserves the previous checkpoint for rollback.

Fresh operator results include exact words 23/23, substring 21/21, prefix 20/22 and suffix only 1/18. In the matched subset, 35 cases improve and one regresses; three remain correct and 18 remain wrong. The suffix weakness is retained as a test finding and is not repaired by training against this now-exposed test.

The new system improves correctness and reduces trainable parameters in this experiment. Its measured prediction latency is higher on the reported test sets. Formal plan search and exact compilation add work, so the correct interpretation is lower adaptation cost and better measured semantics, with a small local response-time cost. A future performance optimization must preserve the same semantic tests.

Failure records remain available at artifacts/plan-comparison/candidate-test.json, alongside baseline cases. There is no test-driven retraining in this run. The 1,057-case audit-test partition in the plan training file is a different set and has not been used to choose this model. The fresh v2 test is now exposed, so a later development cycle needs a new frozen version for independent confirmation.

Website and User Experience

Workspaces and Technology

The primary interface is a React application built with Vite and served from frontend/dist by Starlette. React Router supplies workspaces and navigation. The source uses Lucide icons, Motion interactions and locally packaged fonts. A legacy Streamlit application remains in the repository, but the current local launcher serves the React product. The backend uses Python, PyTorch, Transformers, NLTK, NumPy and optional Manim rendering.

DFA Interaction

The user enters a language description and a binary test string. Learned mode invokes the active evaluated BERT checkpoint or an explicitly selected local checkpoint. Reference demonstrations, intentionally incorrect fixtures and manual DFA JSON are distinct modes with clear provenance. A benchmark example can supply its independent reference; a free-form request generally cannot. The interface shows either verified evidence or an explicitly unverified candidate preview.

The visible graph is generated from actual states and transitions. Accepting-state rings, start markers and edge labels follow the formal object. Simulation highlights the current state and consumed input symbol. Input-only edits recompute the trace for the same machine, while a changed English requirement triggers a new prediction. A suffix-01 requirement produces a three-state minimal machine; suffix-0101 requires five. A longer test word can loop through those five states without creating extra graph nodes.

Analysis panels derive transition tables, state visits, access strings, accepting completions and accepted/rejected examples from the candidate. These facts help inspect behavior but do not prove correspondence to English. A shortest formal counterexample is shown when an independent reference reveals a disagreement. Empty input is handled as epsilon and is not replaced by an arbitrary demonstration string.

Grammar and Compiler Interaction

The grammar workspace displays BERT-produced productions and parse outcomes for the current tokens. Input changes can reuse the same compatible BERT grammar result, then reparse the new sequence. Tree alternatives and derivation playback correspond to actual parser output. Arithmetic mode additionally presents lexemes, a parse structure, AST and symbolic IR. Unsupported grammar output or invalid source is reported without substituting a known fixture.

Presentation Preservation and Responsive Behavior

The current experiment keeps Diagram.jsx, layout.py, scenes.py, renderer.py, styles.css and analysis.css byte-identical to their recorded pre-experiment hashes. Model provenance text is updated where necessary, while the final canvas and playback stay visually consistent. Browser checks exercise desktop and mobile widths, verify that the graph changes with a changed language and that acceptance changes with the test string. The design goal is stable presentation with correct dynamic content.

API, Jobs, Caching and Deployment

Route	Role
GET /api/health	Server readiness, active local model and render capability.
GET /api/catalog	Examples, model choices and grammar availability.
GET /api/research	Measured summaries and experiment metadata.
POST /api/jobs	Submit bounded DFA/CFG/compiler/render work.
GET /api/jobs/{job_id}	Poll progress, result or explicit error.
POST /api/simulate	Recompute a DFA trace from a prior result.
GET /api/videos/{name}	Serve an existing validated cache filename.

The server uses a bounded ThreadPoolExecutor. Local defaults are two workers and at most four running jobs. Result retention is bounded at 128 entries and job history at 256. Request size, description size and input limits prevent accidental unbounded work. Hosted environment variables can reduce workers and job counts. Parsing uses a configurable subprocess timeout rather than allowing an unbounded grammar request to block indefinitely.

Checkpoint dispatch reads config.kind. A semantic_plan checkpoint loads head.pt and its named encoder; direct checkpoints load weights.pt. A shared encoder cache is keyed by resolved path and file revisions. Grammar and plan heads validate their encoder SHA-256 before use. The active-model pointer is explicit; finding an arbitrary folder does not automatically make it the default. Smoke checkpoints are rejected from the learned product route.

Model-generated language_plan metadata is preserved with a DFA result when available. The trace and diagram still consume the ordinary DFA contract. A reference comparison can select a later candidate, so metadata is associated with the selected candidate rather than blindly reporting the first plan. No model state is stored as a mutable last-plan field shared across concurrent requests.

The local launcher builds the frontend when needed and serves port 8501. An existing build can be reused with --skip-build. Docker and Render manifests describe a deployment configuration using the repository and health endpoint. They do not demonstrate that newly generated local weights have been uploaded or remotely activated. Git ignores generated artifacts and model weights, so a remote deployment requires a separate artifact-provisioning process.

The research endpoint adapts the active model's measured original regression result to the existing frontend metric contract and exposes the complete plan comparison separately. This keeps displayed statistics tied to the selected architecture. The UI text identifies the results as a measured regression set rather than continuing to describe a repeatedly inspected set as unseen.

The server treats submitted source and descriptions as data. The arithmetic compiler does not invoke Python eval, shell execution or a generated runtime. The product is a local educational application, not an authenticated multi-tenant inference service. Internet-facing hardening, persistent accounts and distributed job queues are outside the implemented scope.

Grammar Analysis and Checked Derivations

BERT Grammar Decoder

The grammar path uses the evaluated V3 token encoder and a separately trained two-layer cross-attention production decoder. It predicts a finite grammar-token vocabulary and EOS. It does not select a handwritten grammar by matching English text. The head config records vocabulary, sequence length, encoder fingerprint and dataset fingerprint. Invalid grammar output raises an explicit prediction error with the actual candidate productions retained for inspection.

The robust head is trained on 1,652 generated/augmented rows. Its internal validation exactness and generated test reflect canonical production targets, not arbitrary grammar-language equivalence. Known arithmetic forms are intentionally present in training. This makes the compiler workspace reliable for its intended grammar while preventing those fixtures from being advertised as novel held-out discovery.

Parsing and Evidence

The formal CFG component parses grammar text, checks declared productions, validates terminals and runs NLTK Earley parsing within a subprocess budget. Returned trees are bounded in count. A tree is checked by matching each expansion to an actual grammar production and comparing the terminal yield with the input token sequence. Epsilon productions are represented explicitly in derivation handling.

A successful tree proves that the supplied token sequence is generated by that predicted grammar. It does not prove that all strings generated by the grammar satisfy the English requirement. General CFG equivalence is not implemented. Bounded challenge membership checks are useful tests, but they must not be labelled a complete equivalence proof.

Input Dependence and Reuse

A grammar represents a language, so a change from ab to aabb should not require a different grammar for matching counts. The result cache can reuse BERT productions when description, model checkpoint and head/encoder revision match. It reparses the new token sequence and regenerates its tree layout. A new model revision invalidates the reuse. This saves encoder work while preserving correct input-dependent content.

The parser can reveal several derivations for an ambiguous grammar. The interface lets the user inspect alternatives and expansion order, with a clear bound on returned trees. One returned tree does not establish unambiguity. Arithmetic grammar precedence is supplied by productions, while the ambiguous addition fixture intentionally exposes alternative groupings.

Grammar Analysis Outputs

The analysis derives terminal/nonterminal sets, production information, nullable behavior and other inspectable facts from the actual grammar. Tree labels and positions come from returned nodes, not a generic template. Character, token and arithmetic-lexeme modes have different tokenization contracts; a mismatch is an input issue rather than evidence that the model should silently reinterpret source.

Arithmetic Compiler Implementation

Supported Language

The compiler supports symbolic arithmetic expressions with identifiers, numeric literals, parentheses, binary plus/minus/multiplication/division and unary signs. The intended BERT-produced grammar gives multiplication/division higher precedence and binary operators left associativity. Identifiers represent external inputs. The system does not provide statements, declarations, types, loops, functions, control flow, machine code or a runtime interpreter.

Lexing and Concrete Tokens

The lexer records token category, original lexeme and source offsets. It distinguishes identifiers, numbers and punctuation, rejects unsupported characters and preserves decimal literals. It accepts at most 4,000 source characters and 64 arithmetic tokens. Source is treated as data. An invalid sequence such as a+*b fails parsing under the actual predicted grammar rather than being fixed into another expression.

From Parse Tree to AST

The compiler checks that terminal leaves align exactly with the lexed categories. It maps those leaves back to their original identifier or number lexemes. Unary signs create unary AST nodes. Binary operators create ordered children, and parenthesis productions preserve grouping without becoming redundant AST nodes. Unsupported production shapes raise an error instead of assuming a familiar arithmetic grammar.

For (total + 3) * scale - 4 / 2, the AST must preserve the parenthesized addition, multiplication by scale, division of 4 by 2 and the final subtraction order. This is a structural translation from a checked derivation. It is not a claim that the system infers arbitrary compiler semantics from a broad English programming-language specification.

Intermediate Representation and Independent Check

Postorder lowering produces temporaries t0, t1 and subsequent targets. Each instruction records an operator and ordered operands, which may be identifiers, numbers or earlier temporaries. The result identifies the final value. An independent reconstruction checks temporary-before-definition errors, operator arity and operand order, then compares the reconstructed symbolic expression with the AST expression. This catches lost or reordered subexpressions in lowering.

The compiler view displays tokens, AST and instruction sequence derived from the current source. Changing source alters these structures. A simple leaf-only expression can have no intermediate instructions and no expansion animation steps; that is valid behavior, not a missing diagram. Division is kept symbolic, so runtime division-by-zero behavior is outside the implemented guarantee.

Evaluation Scope

Compiler challenge success counts both valid-source acceptance and correct rejection of intentionally invalid source under the desired grammar. The nine-case mixed test is too small to establish broad compiler correctness. Unit tests cover precedence, associativity, unary operators, lexical failure and IR consistency. Further language features require new specifications, BERT supervision and formal checking contracts rather than another generic tree.

Evidence Boundaries and Rendering

Three Different Claims

Structure checked means a supplied or predicted DFA has a complete valid transition table, or a grammar/tree conforms to its explicit contract. Simulated means a trace follows that object for one input. Verified equivalence means the candidate DFA matches an independent reference for every word over the matching alphabet. These claims have different evidence requirements and remain distinct in result records.

An arbitrary English request normally lacks an independent formal reference. The application can show an unverified candidate preview and its trace. That view is useful for inspection and debugging, but the system does not relabel it as exact English correctness. Manual input is also clearly labelled and does not count as BERT prediction.

Sealed Export Objects

The DFA verification gate issues a sealed candidate/reference pair only after exact equivalence. The renderer accepts this evidence object rather than an arbitrary candidate dictionary. It rechecks the seal and equivalence, then reconstructs the requested trace. Tree exports likewise require production and yield checks. The contract prevents a rejected neural prediction from receiving an authoritative-looking verified video.

An incorrect candidate is never replaced by a known constructor output in the learned route. Tests deliberately supply a wrong BERT candidate for a known suffix benchmark and verify that the equivalence failure remains visible. Reference demonstrations are separately labelled, so using them is transparent rather than inflating learned accuracy.

Dynamic Visual Content

Graph layout derives nodes from actual states and edges from actual transitions. Start arrows, self-loops, reciprocal transitions and accepting rings reflect that structure. Trace frames derive the current state from simulation. Parse layout derives nodes and links from actual derivations. AST views derive their nodes from compiler lowering input. The visual system is reusable, while mathematical content is recomputed for each relevant request.

The new experiment records hashes for six presentation files and confirms that all remain unchanged. Thus the final visualization style is preserved. Browser and HTTP checks separately verify the active model, graph counts, changing trace acceptance, grammar tree growth and compiler behavior. A source hash alone would not prove correct runtime integration.

Caching and Export Limits

Video cache keys depend on the validated payload and rendering options. A cache hit reuses a previously created file for the same evidence/input contract. Manim is optional and guarded by availability. Large graphs or trees may exceed readable rendering budgets even when formal checks succeed; the result retains the formal object and reports a visual limit rather than silently drawing an unrelated simplified structure.

Testing and Integration Verification

Formal and Neural-Decoder Tests

New predicate-based compiler tests enumerate binary words through length six and compare simulation against independent Python predicates for conjunction, union, negation, exact counts, count bounds, count modulo and total-length conditions. They cover empty input and overlapping literal patterns. Invalid modulus, invalid patterns, Boolean-as-integer arguments and state-budget violations are rejected.

A decoder test injects ideal logits for a composite plan with literals 00101 and 10. It verifies that learned-copy mechanics preserve leading zeros, respect full literal boundaries and reconstruct the intended typed tree. This tests the decoder independently of neural training quality. Dataset tests verify one partition per canonical language and no training-language overlap with the frozen comparison.

Application Regression Tests

Existing tests cover invalid structure, exact equivalence and shortest counterexamples, strict JSON, unverified manual previews, smoke-checkpoint rejection, no rule fallback, grammar reuse revision checks and compiler failures retaining actual productions. Head-checkpoint integration is checked through the same learned route used by the product. The full suite is rerun after final code changes, with results saved in the verification artifact.

Runtime Checks

The local server is restarted with the updated model dispatcher. HTTP checks exercise asynchronous jobs, independent-reference rejection, valid derivations, compiler tasks and available video export. Browser checks run the real React UI on desktop and mobile. They confirm suffix-01 produces three states, suffix-0101 produces five, a changed input changes acceptance, grammar inputs alter parse-tree size and compiler output follows source.

The six frozen presentation files are SHA-256 compared against their pre-experiment values. These checks protect the requested visualization presentation while model code and provenance copy change. Browser console errors and responsive overflow are checked separately; a passing Python suite alone would not catch a broken frontend contract.

Recorded Verification Status

The final verification record reports: 93 Python tests pass; production frontend build succeeds; HTTP API and video checks pass; dynamic desktop/mobile browser checks pass with no page errors; active model research metrics load; all six frozen visualization hashes match. Results are integration evidence for the supported application paths. They do not convert a finite regression suite into a proof of unrestricted English interpretation.

The research comparison remains separate from software tests. Model semantic scores come from independently generated reference DFAs on the frozen cases. Software tests validate contracts and implementation behavior. Both are necessary, but neither should be substituted for the other in a project presentation.

Reproducibility and Local Operation

Starting the Existing Local Installation

```
.venv/bin/python scripts/run_local.py --port 8501
```

Open http://localhost:8501 after the server reports readiness. The launcher builds the frontend unless --skip-build is supplied. For an already built frontend, use .venv/bin/python scripts/run_local.py --port 8501 --skip-build. The Python environment must include the project dependencies, and Node is needed for rebuilding React. The project source configuration accepts Python 3.10 or later; the recorded local interpreter is Python 3.12.

Data and Model Reproduction

```
.venv/bin/python scripts/download_training_assets.py
.venv/bin/python -m theorybert.data.prepare --help
.venv/bin/python -m theorybert.data.expand --help
.venv/bin/python -m theorybert.data.plans
```

Review the preparation and expansion CLI options before rerunning corpus generation. The plan builder intentionally refuses to overwrite an existing frozen v2 comparison. For an independent later experiment, introduce a new version and freeze it before predictions. Reproducing the plan head requires the exact evaluated V3 encoder, not merely a fresh pretrained BERT download.

```
.venv/bin/python -m theorybert.models.semantic --epochs 60
.venv/bin/python scripts/compare_plan_model.py
.venv/bin/python -m pytest -q
```

The comparison script refuses to overwrite an existing selection. It is an audit boundary, not a command to repeatedly tune the same test. A clean reproduction should use isolated output paths and retain the protocol, dataset fingerprints and encoder artifact. Training is CPU-limited by code and can stop if resource thresholds are exceeded. Do not launch another training job alongside it on this machine.

Selecting or Rolling Back a Model

The active pointer currently selects BERT semantic-plan decoder. The previous direct checkpoint remains artifacts/bert-v3/best. Settings can select a checkpoint explicitly for inspection. A default rollback should update artifacts/active-model.json to an evaluated compatible checkpoint and its measured summary paths. The plan head depends on its recorded V3 encoder; deleting that encoder breaks both plan and grammar loading.

Artifact Availability and Reporting

Model weights, generated datasets and evaluation directories are local and ignored by Git. The repository URL identifies the project, not guaranteed access to this working-tree snapshot. A shareable research release should include a source commit, dependency lock, permitted datasets or deterministic generation instructions, source/license notices, checkpoint checksums and evaluation outputs. The report builder is stored under docs/reports/ and reads the completed local experiment evidence.

The two PDFs can be regenerated with .venv/bin/python docs/reports/build_reports.py after its report dependencies are available. They use embedded Times New Roman fonts and vector architecture diagrams. Their page renderings and text checks are retained as local QA artifacts.

Limitations and Failure Analysis

Language Understanding

The new model remains imperfect on the fresh test despite large gains over V3. Typed validity does not eliminate operator confusion or numeric mistakes. Negation scope, multiple literals, inclusive/exclusive bounds, ambiguous descriptions and unseen terminology can cause incorrect plans. The model has no calibrated guarantee that a high-scoring output matches the English. A valid output can therefore be confidently wrong.

The task is binary and the plan vocabulary is finite. Requests for arbitrary alphabets, nonregular conditions, unrestricted long-count values or complex Boolean nesting are outside demonstrated coverage. The current implementation can still produce a valid in-scope interpretation for an out-of-scope request, so unsupported-intent detection is a research need. Type checks only validate the predicted object; they do not understand whether the request should have been rejected.

Evaluation and Generalization

The comparison uses generated descriptions and one seed. It shares operator families with training and combines an architectural change with more supervision. Fresh test size is 120, with only 57 matched cases. The legacy regression test and retired challenge have been inspected repeatedly. These factors prevent broad claims about unrestricted user language, isolated architectural causality or statistically stable superiority across runs.

The all-case fresh score includes 63 capacity-mismatched cases. It is useful for system coverage, but the matched subset is the appropriate direct-head comparison. Within-five success depends on a reference and cannot be promised for arbitrary free-form English. A general correctness percentage for the entire product would mix incompatible tasks and evidence types.

Grammar and Compiler Scope

Production exactness is sensitive to grammar spelling and order conventions. Alternative equivalent grammars may score differently, and general CFG equivalence is not checked. A successful derivation only proves membership under the predicted productions. Compiler support is limited to symbolic arithmetic and does not include a runtime, optimizer, machine backend, types or full language specification. Correct rejection is counted explicitly in negative compiler fixtures.

Performance and Operational Risks

The head is much smaller to train, but BERT remains the inference backbone and the new plan compiler adds work. The measured response-time cost is real. Resource guards use periodic snapshots and cannot prevent every brief pressure spike or unrelated background disk change. Historical checkpoints already consume substantial local storage, so another experiment should first budget its exact artifact requirements. No automatic deletion policy is introduced.

There has been no controlled educational user study, accessibility audit or internet-scale load test. The interface may help inspect formal behavior, but improved learning outcomes are unmeasured. These are future validation tasks rather than implied benefits of a visually polished system.

Contributions and Development Roadmap

Implemented Contributions

The project combines BERT-based English interpretation with exact formal execution and provenance-aware educational visualization. It provides a direct automata predictor, a learned CFG production decoder, an arithmetic AST/IR path and a new compact plan decoder with learned literal copying. The same final diagram system consumes the formal result, preserving the requested presentation while allowing its content to change correctly.

It also contributes an auditable local workflow: canonical-language splits, pinned public inputs, documented filtering, versioned generated supervision, validation-only deployment gates, retained negative results and resource logs. These are practical project contributions. They build on prior semantic parsing, intermediate representations, formal constraints and classical language theory.

Immediate Next Work

The highest-value next step is a new independent human-authored evaluation set with ambiguous requests, minimal semantic edits and explicit unsupported cases. It should be frozen before model changes and include pairings such as starts versus ends, contains versus avoids, at most versus at least and one literal substitution with leading zeros. Errors should be categorized by operator, numeric argument, literal selection, composition and formal-capacity rejection.

A matched-data experiment should train direct and plan heads with the same eligible supervision and comparable encoder conditions. At least three seeds would reveal variance. Additional controls should isolate learned copying, typed constraints, Boolean compositions and teacher-label choice. The current experiment evaluates the complete approach; these controls would identify which parts produce the gains.

Model and Product Improvements

Confidence calibration and explicit unsupported-request classification should precede a claim of dependable open-ended use. A model-produced plan could be shown in an editable confirmation field, allowing users to correct intent before construction. This would preserve the final visualization while making semantic ambiguity visible. It should be evaluated as a human-in-the-loop system, not counted as unaided model accuracy.

Performance work can profile encoder time, prefix search and formal construction separately. Safe memoization of compatible immutable results may reduce repeated work. Quantization or a smaller encoder should be accepted only after semantic and grammar regression checks. New alphabets, richer grammars and compiler statements require expanded data and new formal contracts, rather than merely increasing a maximum length.

Research Release Readiness

Before external submission, add the actual author/institution block, select a target venue, apply its exact official template and verify page/margin requirements. Release permitted data artifacts or exact generation recipes, cite source notices and archive the trained head plus encoder lineage. The eight-page companion is an IEEE-style project manuscript; this comprehensive report provides the operational evidence and limitations needed to assess it.

Dataset Fingerprints and Exact Source Links

data/challenges/plan-comparison-v2.jsonl
254 rows; validation=134, test=120
SHA-256: 550b42a0566af360e46cea8c6f903c0d824aa805451c198f3549f18b352dc783

data/challenges/toc-compiler-v1.jsonl
165 rows; test=48, development=71, validation=46
SHA-256: ff33ccb8b859220015a605d404f75be78cfc7ca21075557805c0c375ad4bf73b

data/examples/challenges.jsonl
3 rows; challenge=3
SHA-256: 7bfad5cef0b9328c2c0a6927a572f9e17dfdddc2fde2b96d1fadd9650e5aaf0a

data/examples/sample.jsonl
261 rows; train=181, validation=39, test=41
SHA-256: 20220962f7d05b9b38fd24289cfaf07f45240b0bb1d5b0cac11b4fe5bcaa4e76

data/generated/english-dfa-expanded.jsonl
3,207 rows; train=2437, validation=391, test=379
SHA-256: 547e006cc5673ee20a46c6de512510eda7bdbf72d94507cd3a9a3c5b31e898d6

data/generated/english-dfa-robust.jsonl
6,738 rows; train=4924, validation=1173, test=641
SHA-256: bced1069877f875226ae34165c9d0c05f8b5a8fdd7cae7337bf28a4d0534b54a

data/generated/english-dfa-v3.jsonl
6,275 rows; train=4889, validation=745, test=641
SHA-256: a186d971fd113e723dc1ad2ca108d1b8d1c5220710398e0d29248399515e2172

data/generated/english-dfa.jsonl
1,785 rows; train=1363, validation=213, test=209
SHA-256: 7cb45cf3e2b4527aa67f35ee89c12decee6bde58b209d9fb3ab63a98e5bc7415

data/generated/english-plan-v1.jsonl
11,415 rows; train=9187, validation=1171, test=1057
SHA-256: a1f9ab1f907fb52015def4184a383d234ad059cd00d0188430ecb5978b9bf584

data/generated/grammar-robust.jsonl
456 rows; validation=85, train=371
SHA-256: 645f2fe6fc42674d88f843509fe17a3836969187716874f7b2febaae08c18d5e

artifacts/bert-grammar-pilot/dataset.jsonl
1,299 rows; train=999, validation=170, test=130
SHA-256: 2ea73a8b6fdc585d3199d6e9f11d690329367f36dc4dae96085058cb6f05c62a

artifacts/bert-grammar/dataset.jsonl
1,299 rows; train=999, validation=170, test=130
SHA-256: 2ea73a8b6fdc585d3199d6e9f11d690329367f36dc4dae96085058cb6f05c62a

artifacts/grammar-robust/dataset.jsonl
1,652 rows; validation=255, train=1267, test=130
SHA-256: 543c7ac4061d22194a69a493732da4818706e0dae224def6a192e0f08332d676

Exact External Files

kb13-descriptions.txt:
https://raw.githubusercontent.com/nicholaslocascio/deep-regex/096490db7f4b0394fbb46b914cb35a0aa1cba29c/datasets/KB13/src.txt

kb13-regex.txt:
https://raw.githubusercontent.com/nicholaslocascio/deep-regex/096490db7f4b0394fbb46b914cb35a0aa1cba29c/datasets/KB13/targ.txt

LICENSE.md:
https://raw.githubusercontent.com/nicholaslocascio/deep-regex/096490db7f4b0394fbb46b914cb35a0aa1cba29c/datasets/license.md

Bibliography

[1] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding," NAACL-HLT, pp. 4171-4186, 2019.
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https://aclanthology.org/2020.findings-emnlp.225/

[9] N. Locascio, DeepRegex code and KB13 dataset mirror, revision 096490db7f4b0394fbb46b914cb35a0aa1cba29c, MIT dataset license.
https://github.com/nicholaslocascio/deep-regex/tree/096490db7f4b0394fbb46b914cb35a0aa1cba29c/datasets/KB13

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https://www.nltk.org/api/nltk.parse.earleychart.html

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https://www.cs.cornell.edu/courses/cs4120/2023sp/notes/leximpl/

[13] TheoryBERT project repository, local implementation and recorded experiments. The local working tree contains changes that may not yet be published.
https://github.com/samyaksrijan/theoryBERT

Public sources checked on 29 October 2026. Original research is cited for context and method inspiration. All project experiment figures come from the local manifests and completed run artifacts. Repository links do not assert that current local changes or large model files have been published.

Additional Local Evidence

artifacts/plan-promotion-protocol.json; artifacts/plan-comparison/selection.json; artifacts/plan-comparison/summary.json; artifacts/bert-plan/best/summary.json; artifacts/plan-verification.json; artifacts/challenge-comparison.json; artifacts/report-data-inventory.json; artifacts/plan-visual-check.json.

Implementation Map and Glossary

Source location	Responsibility
src/theorybert/models/theorybert.py	Pretrained encoder, adapted features and direct DFA heads.
src/theorybert/models/semantic.py	Plan head, literal pointers, decoding and guarded training.
src/theorybert/models/predictor.py	Checkpoint dispatch and compatible encoder injection.
src/theorybert/models/grammar.py	Production-token decoder, grammar corpus and training.
src/theorybert/formal/plan.py	Typed plan validation and exact DFA construction.
src/theorybert/formal/dfa.py	Validation, trace, minimization and equivalence.
src/theorybert/formal/cfg.py	Bounded parsing and derivation checks.
src/theorybert/formal/compiler.py	Arithmetic lexing, AST and checked symbolic IR.
src/theorybert/formal/analysis.py	Machine/grammar facts derived from current objects.
src/theorybert/web.py	API jobs, active artifacts, cache and provenance.
frontend/src/App.jsx	Workspaces, inputs, evidence, playback and settings.
frontend/src/components/Diagram.jsx	Reusable graph/tree visual components.
src/theorybert/rendering/	Geometry, sealed rendering payloads and Manim scenes.
src/theorybert/data/	Generation, import, augmentation, challenges and plans.
scripts/compare_plan_model.py	Frozen validation selection and later test evaluation.

Recorded Local Technology Versions

Technology	Version	Technology	Version
Python	3.12.4	torch	2.14.1
transformers	4.57.6	starlette	1.9.0
uvicorn	0.54.0	nltk	3.10.4
numpy	2.5.4	manim	0.22.0
streamlit	1.66.1	react	19.3.1
react-router-dom	7.18.4	vite	8.3.5
motion	14.6.0	lucide-react	1.61.0

Glossary

DFA: deterministic finite automaton. CFG: context-free grammar. AST: abstract syntax tree. IR: intermediate representation. EOS: end-of-sequence marker. Canonical language identity: minimal reachable DFA renamed in fixed BFS order. First-candidate equivalence: exact semantic correctness before reference-assisted search. Within-five: a correct candidate found with an independent reference. Plan: model-produced typed regular-language expression. Frozen encoder: weights reused without gradient updates. Preview: inspectable candidate without independent equivalence evidence. Sealed evidence: a verified formal object authorized for the corresponding rendering contract.

The companion manuscript presents the research argument, while this report supplies the detailed implementation and operational record. The dataset archive supplies local supervision and provenance. Reproducing the learned outputs also requires the fingerprinted evaluated encoder and compatible trained heads.

Expanded Review of the Supplied Literature

Attention, Pretraining and Adaptation

Vaswani et al. [14] provide the attention-based Transformer foundation. RoBERTa [15] emphasizes careful control of pretraining data and optimization. Together they support treating architecture, supervision and training condition as separate experimental variables. TheoryBERT uses the original base uncased BERT lineage, not RoBERTa, and its direct-versus-plan comparison changes both representation and supervised data.

BioBERT [16] demonstrates domain-specific continued pretraining in biomedicine. The more recent ModernBERT legal adaptation study [24] supplies another domain-adaptation example. These works motivate domain sensitivity but do not imply that biomedical or legal data would improve automata reasoning. Neither corpus nor checkpoint family is used in the local experiments. A formal-language continued-pretraining study would need its own corpus, cost budget and held-out evaluation.

Vocabulary, Distillation and Compute

The publisher-indexed Efficient Token Augmented Fine-Tuning record [19] identifies token augmentation as an alternative to expensive continued pretraining. Its full experimental text was not accessible in this research pass, so the report does not infer its initialization recipe or reproduce its performance claims. The supplied venue acronym is corrected to CSDE. This distinction matters because a valid citation is not evidence that a method was implemented here.

Ryu and Lee [20] study adversarial adaptation with distillation for domain shift. Linder's KTH thesis [21] studies vocabulary expansion through self-distillation across tokenizations. These are plausible future controls for preserving old behavior while learning domain terms. The completed plan run uses supervised token and literal-pointer losses, not teacher-distribution distillation. Its efficiency result comes from a small head over frozen features.

Formal Languages and Semantic Parsing

Deletang et al. [17] study neural generalization across formal-language task classes. Strobl et al. [18] organize transformer-expressivity results under their assumptions. Their relevance is a warning: success on finite generated examples does not establish unrestricted recognition or compositional generalization. TheoryBERT delegates execution to classical formal algorithms after a bounded learned interpretation.

Locascio et al. [3] remain directly relevant to natural-language regex generation and the public-data lineage. Xu et al. [25] provide a separate semantic-parsing example over Wikidata. The latter uses another task, data source and model family. TheoryBERT does not use WikiWebQuestions, Wikidata, LLaMA or GPT as a hidden prediction path.

Peripheral Context and Scope

The visual-language study by Alper et al. [22] concerns implicit visual knowledge in language representations. It does not validate drawing DFA graphs. CyMapNER [23] concerns domain-specific named entity recognition, not automata construction. Both are verified references and useful broader context, but neither supplies implementation evidence for this prototype. None of the extra literature datasets is included in the local training totals.

Supplemental Bibliography

These numbers continue the comprehensive report bibliography. The eight-page manuscript independently numbers its selected sources by first citation order.

[14] A. Vaswani et al., "Attention Is All You Need," Advances in Neural Information Processing Systems, vol. 30, 2017.
https://proceedings.neurips.cc/paper_files/paper/2017/hash/3f5ee243547dee91fbd053c1c4a845aa-Abstract.html

[15] Y. Liu et al., "RoBERTa: A Robustly Optimized BERT Pretraining Approach," arXiv:1907.11692, 2019.
https://arxiv.org/abs/1907.11692

[16] J. Lee et al., "BioBERT: a pre-trained biomedical language representation model for biomedical text mining," Bioinformatics, vol. 36, no. 4, pp. 1234-1240, 2020.
https://academic.oup.com/bioinformatics/article/36/4/1234/5566506

[17] G. Deletang et al., "Neural Networks and the Chomsky Hierarchy," ICLR, 2023; arXiv:2207.02098.
https://arxiv.org/abs/2207.02098

[18] L. Strobl, W. Merrill, G. Weiss, D. Chiang, and D. Angluin, "What Formal Languages Can Transformers Express? A Survey," TACL, vol. 12, pp. 543-561, 2024.
https://aclanthology.org/2024.tacl-1.30/

[19] L. J. Weber, K. J. Ramalingam, M. Beyer, A. Kirchheim, and A. Zimmermann, "Efficient Token Augmented Fine-Tuning," IEEE Asia-Pacific Conference on Computer Science and Data Engineering (CSDE), 2023, doi:10.1109/CSDE59766.2023.10487678.
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[20] M. Ryu and K. Lee, "Knowledge Distillation for BERT Unsupervised Domain Adaptation," arXiv:2010.11478, 2020.
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[21] M. R. Linder, "Vocabulary Expansion of Large Language Models via Self-Distillation," M.S. thesis, KTH Royal Institute of Technology, Jan. 2026; arXiv:2508.15807v2.
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[22] M. Alper, M. Fiman, and H. Averbuch-Elor, "Is BERT Blind? Exploring the Effect of Vision-and-Language Pretraining on Visual Language Understanding," CVPR, 2023.
https://openaccess.thecvf.com/content/CVPR2023/papers/Alper_Is_BERT_Blind_Exploring_the_Effect_of_Vision-and-Language_Pretraining_on_CVPR_2023_paper.pdf

[23] J. Zhang, F. Shi, C. Xu, Y. Li, X. Yin, and M. Ge, "CyMapNER: a named entity recognition model for cyberspace surveying and mapping domain," Cybersecurity, vol. 9, art. 146, 2026.
https://link.springer.com/article/10.1186/s42400-026-00578-3

[24] D. Stammbach and P. Henderson, "Legal Domain Adaptation of Modern BERT Models," arXiv:2606.28538, 2026.
https://arxiv.org/abs/2606.28538

[25] S. Xu et al., "Fine-tuned LLMs Know More, Hallucinate Less with Few-Shot Sequence-to-Sequence Semantic Parsing over Wikidata," EMNLP, pp. 5778-5791, 2023.
https://aclanthology.org/2023.emnlp-main.353/

TAFT metadata and abstract were checked in the publisher-indexed proceedings; the full IEEE paper was inaccessible. The audit on the following pages distinguishes verified metadata, corrected details and project relevance. Literature-only datasets and models are not project training inputs.

Reference Audit: Supplied Items 1-8

Supplied item numbers refer to the bibliography provided by the user. Report reference numbers refer to the comprehensive report. Verification checks identity and publication metadata; it does not imply a reproduced experiment.

Supplied [1] / report [14]: Verified
A. Vaswani et al., "Attention Is All You Need," Advances in Neural Information Processing Systems, vol. 30, 2017.
The official NeurIPS proceedings confirm the 2017 Transformer paper. Used as the attention foundation; its translation data are not local project datasets. Primary record

Supplied [2] / report [1]: Verified
J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding," NAACL-HLT, pp. 4171-4186, 2019.
NAACL-HLT 2019 record confirms authors and title. This is the actual encoder lineage used through the pinned base uncased artifact. Primary record

Supplied [3] / report [15]: Verified, contextual
Y. Liu et al., "RoBERTa: A Robustly Optimized BERT Pretraining Approach," arXiv:1907.11692, 2019.
The 2019 arXiv record is correct. RoBERTa is discussed as pretraining-control context; no RoBERTa checkpoint or corpus is used here. Primary record

Supplied [4] / report [4]: Verified
W. Tai, H. T. Kung, X. Dong, M. Comiter, and C.-F. Kuo, "exBERT: Extending Pre-trained Models with Domain-specific Vocabulary Under Constrained Training Resources," Findings of EMNLP, pp. 1433-1439, 2020.
Findings of EMNLP 2020 confirms the citation. The project extension is exBERT-inspired and is not an exact architecture or biomedical-training reproduction. Primary record

Supplied [5] / report [16]: Verified, full title restored
J. Lee et al., "BioBERT: a pre-trained biomedical language representation model for biomedical text mining," Bioinformatics, vol. 36, no. 4, pp. 1234-1240, 2020.
The full title ends with 'for biomedical text mining'. The issue year is 2020 although advance online publication was September 2019. No BioBERT weights or biomedical corpora are used. Primary record

Supplied [6] / report [20]: Verified as the 2020 preprint
M. Ryu and K. Lee, "Knowledge Distillation for BERT Unsupervised Domain Adaptation," arXiv:2010.11478, 2020.
The arXiv version lists Minho Ryu and Kichun Lee. A later 2022 journal version also lists Geonseok Lee; these are different bibliographic versions. Distillation is not implemented in this project. Primary record

Supplied [7] / report [19]: Venue acronym corrected; abstract-level verification
L. J. Weber, K. J. Ramalingam, M. Beyer, A. Kirchheim, and A. Zimmermann, "Efficient Token Augmented Fine-Tuning," IEEE Asia-Pacific Conference on Computer Science and Data Engineering (CSDE), 2023, doi:10.1109/CSDE59766.2023.10487678.
The publisher index identifies IEEE CSDE 2023, not APSCDE. Title and five authors match. DOI is 10.1109/CSDE59766.2023.10487678. Full paper access was unavailable, so detailed experimental claims are not used. Primary record

Supplied [8] / report [21]: Thesis verified; version clarified
M. R. Linder, "Vocabulary Expansion of Large Language Models via Self-Distillation," M.S. thesis, KTH Royal Institute of Technology, Jan. 2026; arXiv:2508.15807v2.
The PDF cover identifies KTH, the exact supplied thesis title and 14 January 2026. The arXiv landing title adds Kullback-Leibler-Based; v1 is August 2025 and v2 is January 2026. No self-distillation run is claimed here. Primary record

Checked 29 October 2026. The authors of cited works are not the authors of this project. The project author/institution block remains omitted because names were not supplied.

Reference Audit: Supplied Items 9-15

Supplied item numbers refer to the bibliography provided by the user. Report reference numbers refer to the comprehensive report. Verification checks identity and publication metadata; it does not imply a reproduced experiment.

Supplied [9] / report [22]: Verified, peripheral
M. Alper, M. Fiman, and H. Averbuch-Elor, "Is BERT Blind? Exploring the Effect of Vision-and-Language Pretraining on Visual Language Understanding," CVPR, 2023.
The CVPR author paper confirms the title and authors. It studies visual-language knowledge, not the correctness of generated DFA diagrams. It is contextual only; this project has no vision-language pretraining. Primary record

Supplied [10] / report [23]: Verified, title expanded
J. Zhang, F. Shi, C. Xu, Y. Li, X. Yin, and M. Ge, "CyMapNER: a named entity recognition model for cyberspace surveying and mapping domain," Cybersecurity, vol. 9, art. 146, 2026.
The publisher gives 'a named entity recognition model', Cybersecurity volume 9, article 146, published 2 April 2026. It concerns cyberspace NER, not automata inference. Its datasets are not project inputs. Primary record

Supplied [11] / report [24]: Verified 2026 preprint
D. Stammbach and P. Henderson, "Legal Domain Adaptation of Modern BERT Models," arXiv:2606.28538, 2026.
arXiv:2606.28538 confirms Stammbach and Henderson. It adapts ModernBERT in law, whereas this project uses original base uncased BERT and supervised formal outputs. Legal corpora and checkpoints are not used. Primary record

Supplied [12] / report [3]: Verified; proceedings version used
N. Locascio, K. Narasimhan, E. DeLeon, N. Kushman, and R. Barzilay, "Neural Generation of Regular Expressions from Natural Language with Minimal Domain Knowledge," EMNLP, pp. 1918-1923, 2016.
EMNLP 2016, pages 1918-1923, DOI 10.18653/v1/D16-1197. This is the directly relevant natural-language regex paper and repository lineage for the filtered KB13 mirror. Primary record

Supplied [13] / report [17]: Verified
G. Deletang et al., "Neural Networks and the Chomsky Hierarchy," ICLR, 2023; arXiv:2207.02098.
The author arXiv PDF states ICLR 2023; its initial preprint is 2022. Its generalization study motivates careful formal-task boundaries, not a claim that this prototype reproduces the study. Primary record

Supplied [14] / report [18]: Verified
L. Strobl, W. Merrill, G. Weiss, D. Chiang, and D. Angluin, "What Formal Languages Can Transformers Express? A Survey," TACL, vol. 12, pp. 543-561, 2024.
TACL volume 12, pages 543-561, 2024, DOI 10.1162/tacl_a_00663. The survey highlights architecture-dependent assumptions; it does not prove that this BERT system solves all formal languages. Primary record

Supplied [15] / report [25]: Verified, contextual
S. Xu et al., "Fine-tuned LLMs Know More, Hallucinate Less with Few-Shot Sequence-to-Sequence Semantic Parsing over Wikidata," EMNLP, pp. 5778-5791, 2023.
ACL confirms EMNLP 2023, pages 5778-5791, DOI 10.18653/v1/2023.emnlp-main.353. Its Wikidata semantic parser is relevant background; no WikiWebQuestions/Wikidata data or LLaMA/GPT route is used here. Primary record

Checked 29 October 2026. The authors of cited works are not the authors of this project. The project author/institution block remains omitted because names were not supplied.